Lecture 3.1: Risk, Return, and Diversification

Risk, Return, and Diversification

- Investors prefer more return and less risk
- We will show that diversification enables enables investors to decrease risk without sacrificing return
- We need to quantify "risk," "return," and "diversification" so that they can be useful concepts in real-world applications

Computing Returns

- We will let $R_{s,t}$ denote the random return of stock s on date t

$$R_{s,t} = \frac{P_{s,t} + D_{s,t} - P_{s,t-1}}{P_{s,t-1}}$$

- We will focus on monthly returns, although you can just as well think of hourly, daily, weekly, monthly, yearly returns
- Example: On 2020-07-31, Apple (AAPL) traded for 106.0643 USD. On 2020-08-31, Apple traded for 129.04 USD and paid a dividend of 0.82 USD. What was Apple's return during the month of August?

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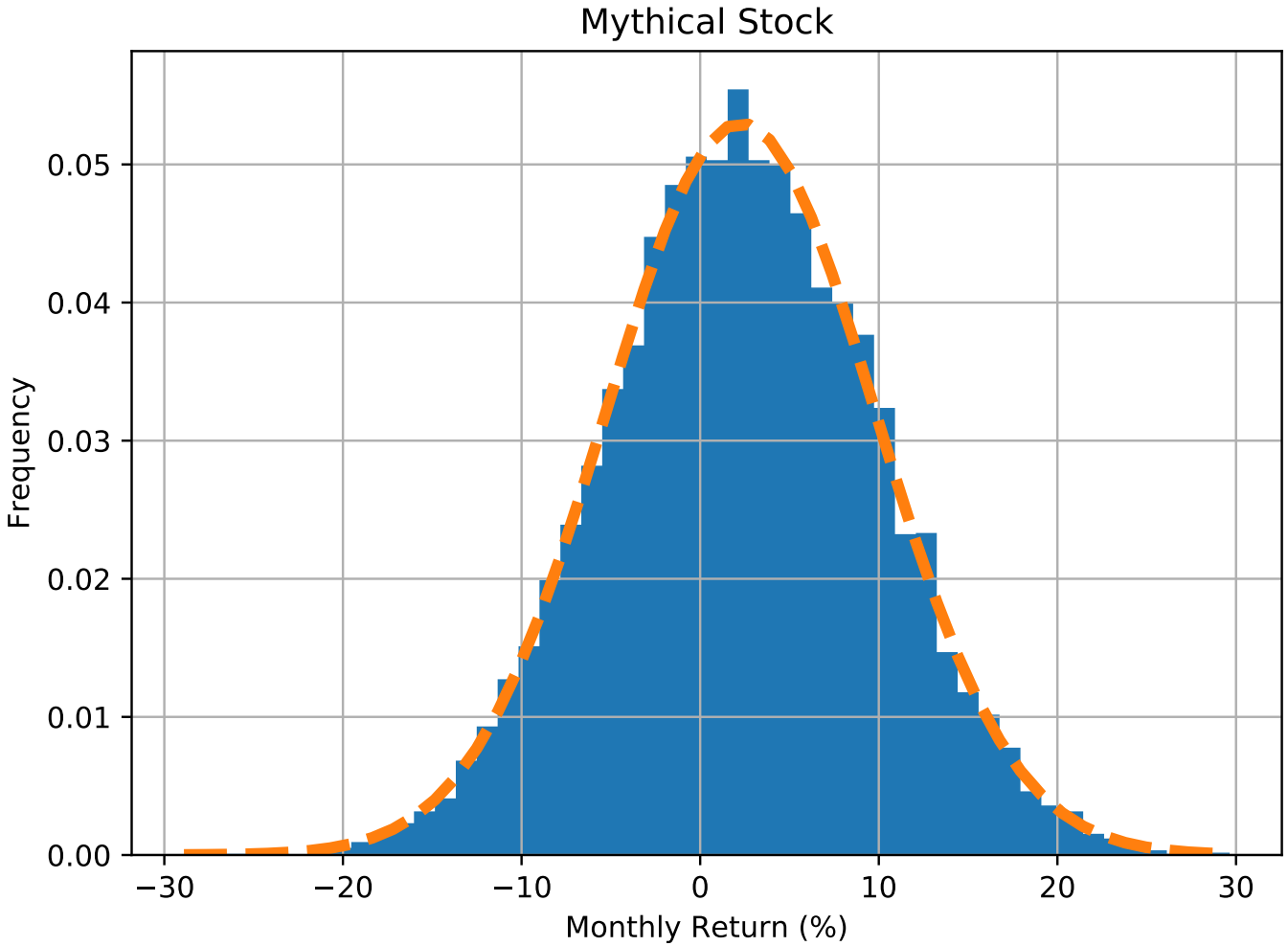
$$\begin{aligned} R_{\text{AAPL,August}} &= \frac{P_{\text{AAPL,August}} + D_{\text{AAPL,August}} - P_{\text{AAPL,July}}}{P_{\text{AAPL,July}}} && (1) \\ &= \frac{129.04 \text{ USD} + 0.82 \text{ USD} - 106.0643 \text{ USD}}{106.0643 \text{ USD}} && (2) \\ &= 22.44\% && (3) \end{aligned}$$

The Normal Distribution

- We assume that returns are normally distributed. Under this assumption, the returns of a portfolio are also normally distributed
- The returns of a portfolio are characterized by three attributes of its constituent stocks:
  1. the expected return of each stock;
  2. the volatility of each stock;
  3. the correlation between each pair of stocks.

The Normal Distribution



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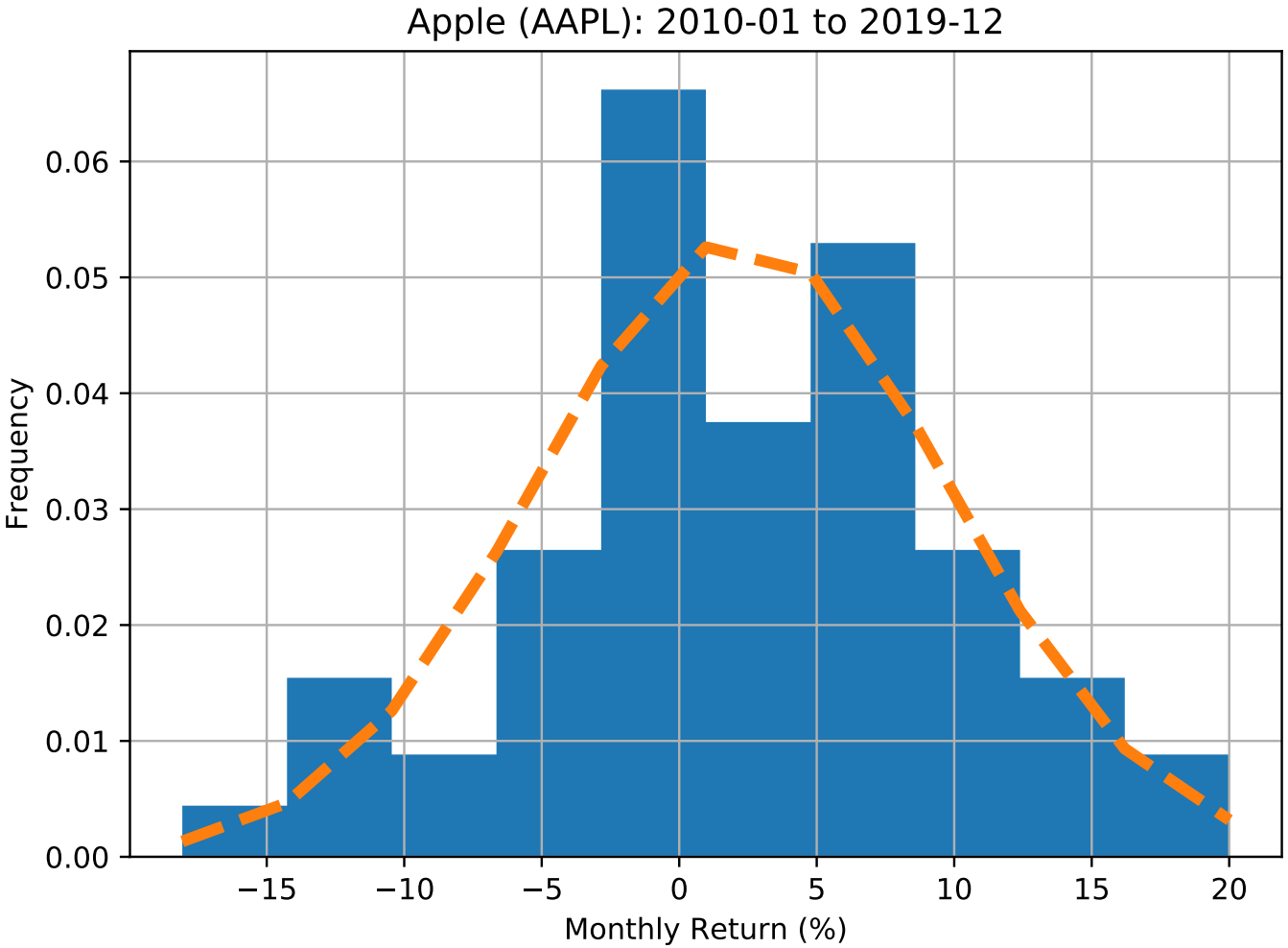
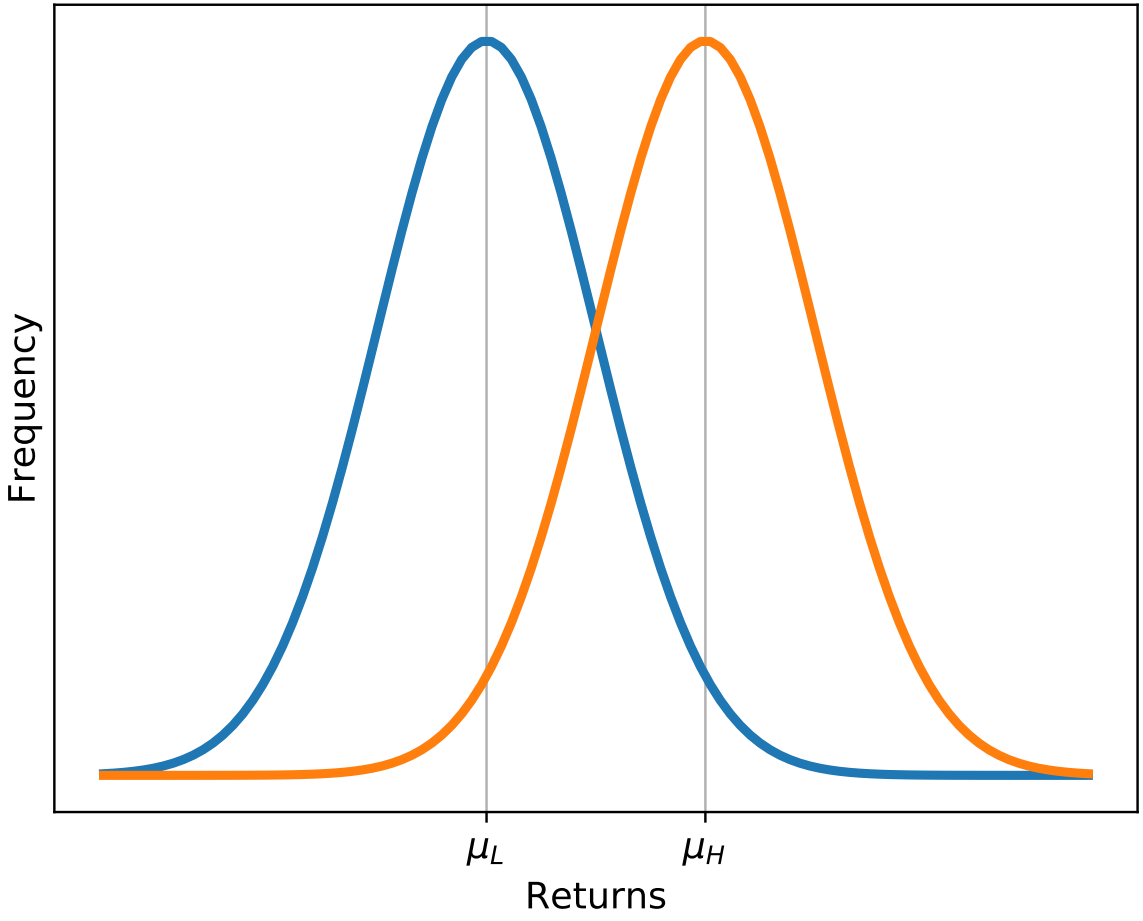
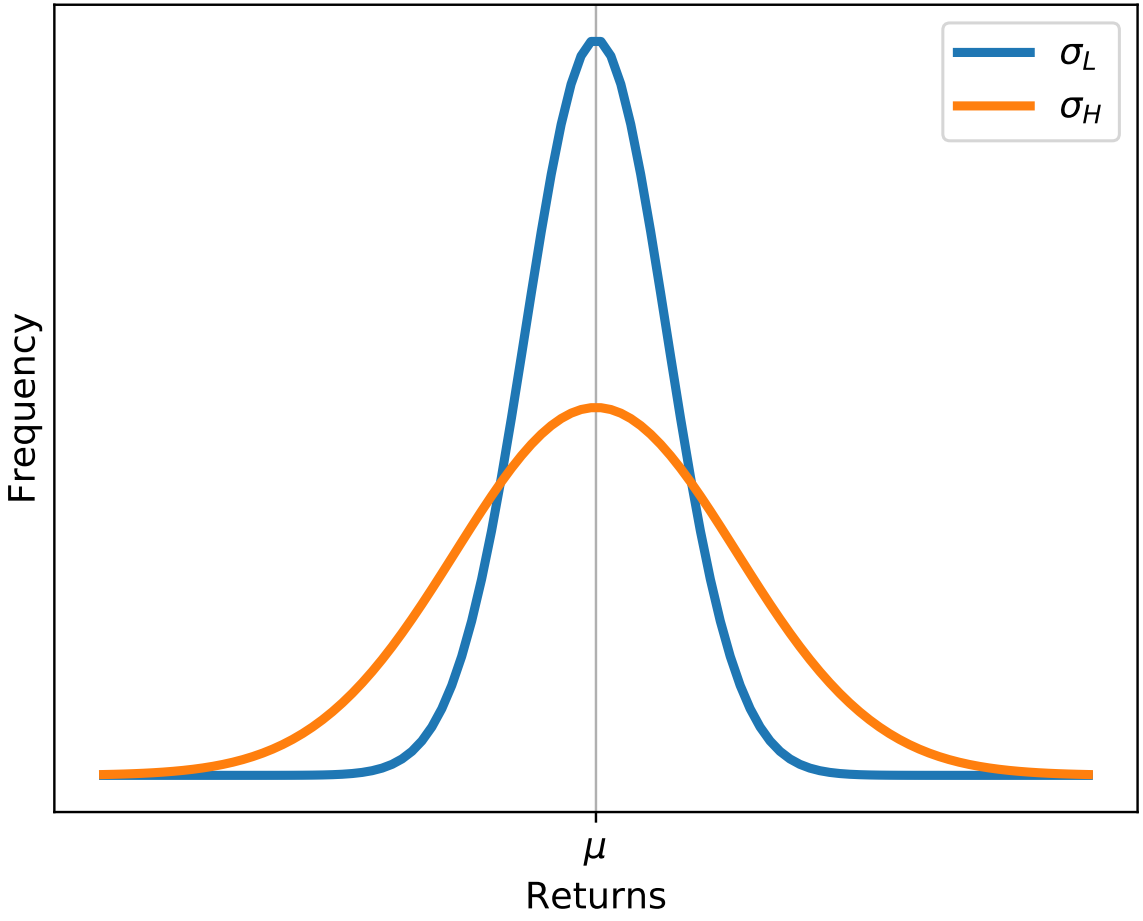


Different Means



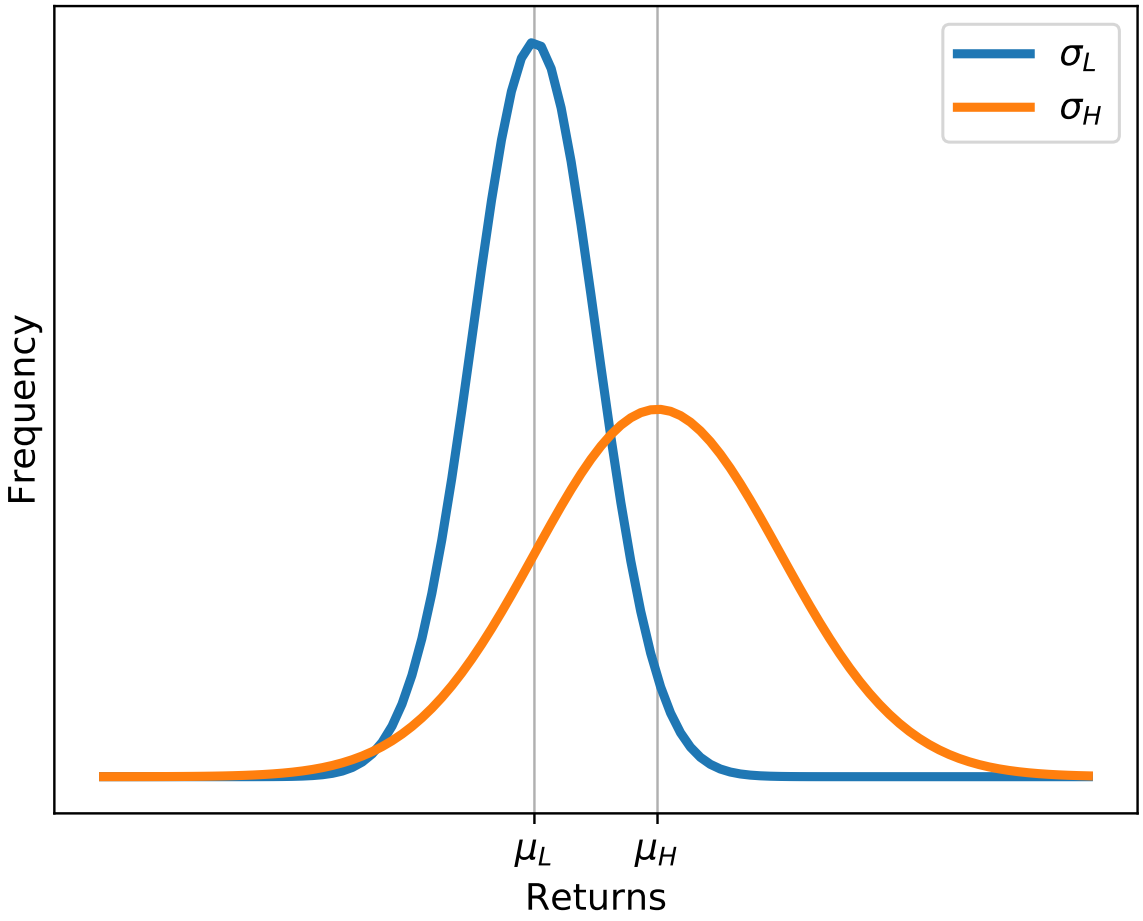
The orange distribution has a higher mean ($\mu_H > \mu_L$)

Different Standard Deviations



The orange distribution has a higher standard deviation ($\sigma_H > \sigma_L$)

Different Means and Standard Deviations



The orange distribution has a higher mean and standard deviation

Expected Return

- We denoted by $\mathrm{E}[R_s]$ the mean, or expectation of R_s
- It is just a <u>number</u>
- It is the <u>expected return</u>
- The mean is expressed in "%"
- Our <u>estimate</u> of the mean return for stock s is

$$\mu_s = \frac{1}{N} \sum_{t=1}^{N} R_{s,t}$$

where $R_{s,t}$ are the <u>realized</u> returns

Volatility

- We denoted by $\mathrm{Var}[R_s]$ the variance of R_s
- The variance is used to construct the standard deviation
- The variance does <u>not</u> have units
- Our <u>estimate</u> of the variance of stock s is

$$\sigma_s^2 = \frac{1}{N-1} \sum_{t=1}^{N} (R_{s,t} - \mu_s)^2$$

- The "$N-1$" makes the estimator unbiased
- We denote by $\mathrm{SD}[R_s]$ the standard deviation or <u>volatility</u> of R_s
- It is our measure of <u>risk</u>
- The standard deviation is $\mathrm{SD}[R_s] = \sqrt{\mathrm{Var}[R_s]}$
- It is expressed in "%"

Covariance

- Let's think about two stocks, A and B
- We denote by $\text{Cov}[R_A, R_B]$ the covariance between the returns on A and B
- The covariance is used to construct the correlation
- The covariance does <u>not</u> have units
- Our <u>estimate</u> of the covariance is

$$\sigma_{A,B} = \frac{1}{N-1} \sum_{t=1}^{N} (R_{A,t} - \mu_A)(R_{B,t} - \mu_B)$$

Correlation

- We denote by $\text{Corr}[R_A, R_B]$ the correlation between the returns of A and B
- It is given by

$$\text{Corr}[R_A, R_B] = \frac{\text{Cov}[R_A, R_B]}{\text{SD}[R_A]\text{SD}[R_B]}$$

- The correlation is between -1 and $+1$, with a correlation of -1 meaning perfectly negatively correlated (A goes up, B goes down), a correlation of 0 meaning they are not correlated at all (A's return doesn't tell us anything about B's return), and a correlation of $+1$ meaning perfectly positively correlated (A goes up, B goes up)

Portfolio of Two Stocks

- We are going to focus on portfolios with two stocks
- All of the computations that follow can be done with an arbitrary number of stocks — we just need some linear algebra
- Consider a portfolio with x_A in stock A and x_B in stock B
- x_A and x_B must sum to one: $x_A + x_B = 1$
- $x > 1$ means we are borrowing (or financing with short)
- $x < 0$ means we are shorting
- The random return of the portfolio is

$$R_P = x_A R_A + x_B R_B$$

Mean and Volatility

- The expected return of the portfolio is

$$\mathrm{E}[R_P] = x_A \mathrm{E}[R_A] + x_B \mathrm{E}[R_B]$$

- The variance of the portfolio is

$$\begin{aligned}\mathrm{Var}[R_P] &= x_A^2 \mathrm{Var}[R_A] + x_B^2 \mathrm{Var}[R_B] + 2x_A x_B \mathrm{Cov}[R_A, R_B] \\ &= (x_A \mathrm{SD}[R_A])^2 + (x_B \mathrm{SD}[R_B])^2 + 2x_A x_B \mathrm{SD}[R_A]\mathrm{SD}[R_B]\mathrm{Corr}[R_A, R_B]\end{aligned}$$

- Therefore, the volatility of the portfolio is

$$\mathrm{SD}[R_P] = \sqrt{\mathrm{Var}[R_P]}$$

Apple and P&G

- What are the expected returns and volatilities of Apple and P&G? How do they compare in terms of risk and return?
- What is the correlation between the returns?
- What is the expected return of a portfolio that is split evenly between Apple and P&G? What is the volatility of the portfolio?

Apple and P&G

- What are the expected returns and volatilities of Apple and P&G? How do they compare in terms of risk and return?
- What is the correlation between the returns?
- What is the expected return of a portfolio that is split evenly between Apple and P&G? What is the volatility of the portfolio?
- Expected returns are 3.56% and 1.09% (resp.) and volatilities are 12.21% and 5.21% (resp.)
- -0.0541 (weird!)
- 2.33% and 6.51%

Risk Premia

Risk-Free Asset

- The risk-free rate is the rate of return that can be earned with certainty
- The risk-free asset is the asset that yields a risk-free return
- We take US Treasuries to be risk-free assets and Treasury yields to be the risk-free return
- Let's call the risk-free rate R_F

Risk Premia

- The risk premium is the expected excess return over the risk-free rate
$$\mathrm{RP}[R_P] = \mathrm{E}[R_P] - R_F$$
- It is the return we earn for taking risk: higher risk stocks will generally have higher risk premia, safer stocks will generally have lower risk premia
- If the portfolio P is risk-free, its risk-premium will be zero

Risk Premia

The mean monthly return on the market from 1929 to 2019 was 0.9379%. The mean monthly return on US T-bills was 0.2732%. What was the market risk premium over this period?

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The monthly market risk premium was

$$\mathrm{RP}[R_{\mathrm{Market}}] = 0.9379\% - 0.2732\% = 0.6647\%$$

This is how much the market had to compensate investors for the risk they were taking.

Sharpe Ratio

Sharpe Ratios

- The Sharpe ratio is the ratio of a portfolio's risk premium to its standard deviation:

$$\mathrm{SR}[R_P] = \frac{\mathrm{RP}[R_P]}{\mathrm{SD}[R_P]} = \frac{\mathrm{E}[R_P] - R_F}{\mathrm{SD}[R_P]}$$

  where $\mathrm{E}[R_P]$ and $\mathrm{SD}[R_P]$ are the mean and standard deviation of the return on the portfolio respectively
- It is the return we earn for each unit of risk we take

Sharpe Ratios

The mean monthly standard deviation of market returns from 1929 to 2019 was 5.3324%. What was the market Sharpe ratio over this period?

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$$\mathrm{SR}[R_{\mathrm{Market}}] = \frac{.6647\%}{5.3324\%} = 0.1247$$

Diversification

Moral #1

Suppose that $R_F = 0$. Consider two stocks, A and B:

Stock	$E[R]$	$SD[R]$	$SR[R]$
A	5%	20%	0.25
B	4%	20%	0.20

Stock B's Sharpe ratio is smaller than that of stock A. Why would anyone <u>ever</u> buy stock B?

Moral #1

Suppose that A and B have a correlation of 0.5. Then

$$\begin{aligned}\mathrm{Var}[R_P] &= x_A^2 \mathrm{SD}[R_A]^2 + x_B^2 \mathrm{SD}[R_B]^2 \\ &\qquad + 2x_A x_B \mathrm{SD}[R_A]\mathrm{SD}[R_B]\mathrm{Corr}[R_A, R_B] \\ &= (.5)^2(.2)^2 + (.5)^2(.2)^2 + 2(.5)(.5)(.2)(.2)(.5) \\ &= .03\end{aligned}$$

and hence $\mathrm{SD}[R_P] = 17.3205\%$.

Moral #1

Take a portfolio-based perspective:

x_A	x_B	$E[R]$	$SD[R]$	$SR[R]$
100%	0%	5%	20%	0.25
0%	100%	4%	20%	0.20
50%	50%	4.5%	17.3205%	0.2598

The new portfolio returns less than stock A on average (4.5% instead of 5%), but has less volatility (17.3205% instead of 20%).

Moral: a low-Sharpe ratio stock can increase the portfolio-Sharpe ratio above the Sharpe ratios of any of the constituent stocks

Moral #2

Suppose that $R_F = 0$. Consider three stocks: A, B, and C. For each stock, $E[R] = 5\%$ and $SD[R] = 20\%$. Therefore, all stocks have a Sharpe ratio of 0.25. The correlations are

$$\text{Corr}[R_A, R_B] = 0.75$$
$$\text{Corr}[R_B, R_C] = 0.50$$
$$\text{Corr}[R_A, R_C] = 0.25$$

Suppose you can only pick two of the three stocks. Which pair would you choose? Suppose that you invest equally in the two stocks.

Moral #2

Let's compute the Sharpe ratios of each of the three portfolios.

x_A	x_B	x_C	Corr	E[R]	SD[R]	SR[R]
50%	50%	0%	0.75	5%	18.7083%	0.2673
0%	50%	50%	0.50	5%	17.3205%	0.2887
50%	0%	50%	0.25	5%	15.8114%	0.3162

We should choose the portfolio with stocks A and C.

Moral: By choosing less correlated stocks, we can increase the portfolio-Sharpe ratio <u>without</u> sacrificing return.

Moral #3

Suppose that $R_F = 0$. Consider three stocks: A, B, and C. For each stock, $E[R] = 5\%$ and $SD[R] = 20\%$. Therefore, all stocks have a Sharpe ratio of 0.25. The correlations are

$$\text{Corr}[R_A, R_B] = +1$$
$$\text{Corr}[R_A, R_C] = -1$$

What are the expected returns, volatilities, and Sharpe ratios of each of the pairs?

Moral #3

Let's compute the Sharpe ratios of each of the three portfolios.

x_A	x_B	x_C	Corr	E[R]	SD[R]	SR[R]
50%	50%	0%	$+1$	5%	20%	.25
50%	0%	50%	-1	5%	0%	?

We should choose the portfolio with stocks A and C. This portfolio is perfectly hedged. It is equivalent to a risk-free asset.

Moral: Perfectly positively correlated stocks behave like one stock, while perfectly negatively correlated stocks behave like the risk-free asset

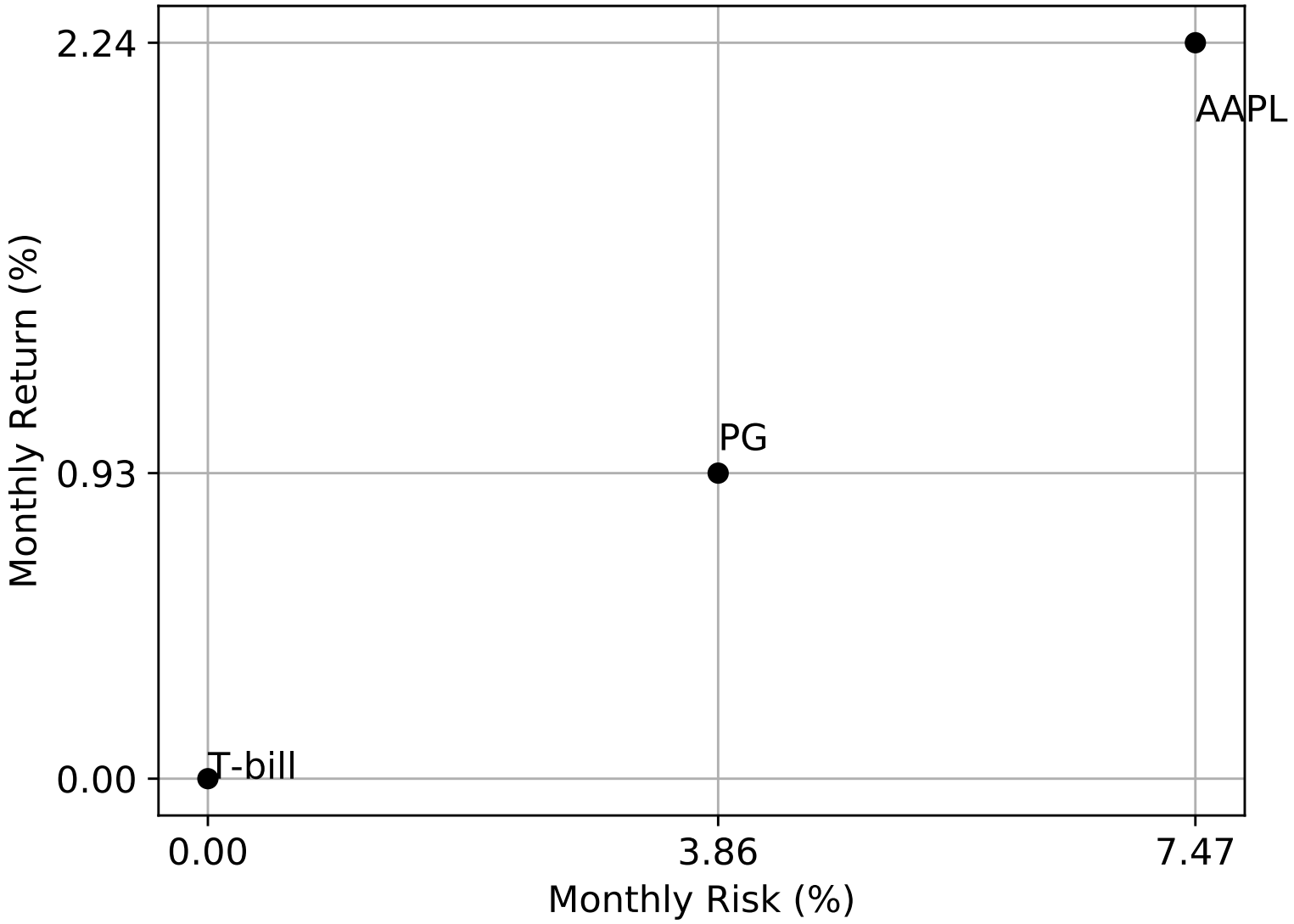
Risk-Return Space

- Let's look at Apple (AAPL) and Proctor & Gamble (PG)
- Their returns have a correlation of 0.3162

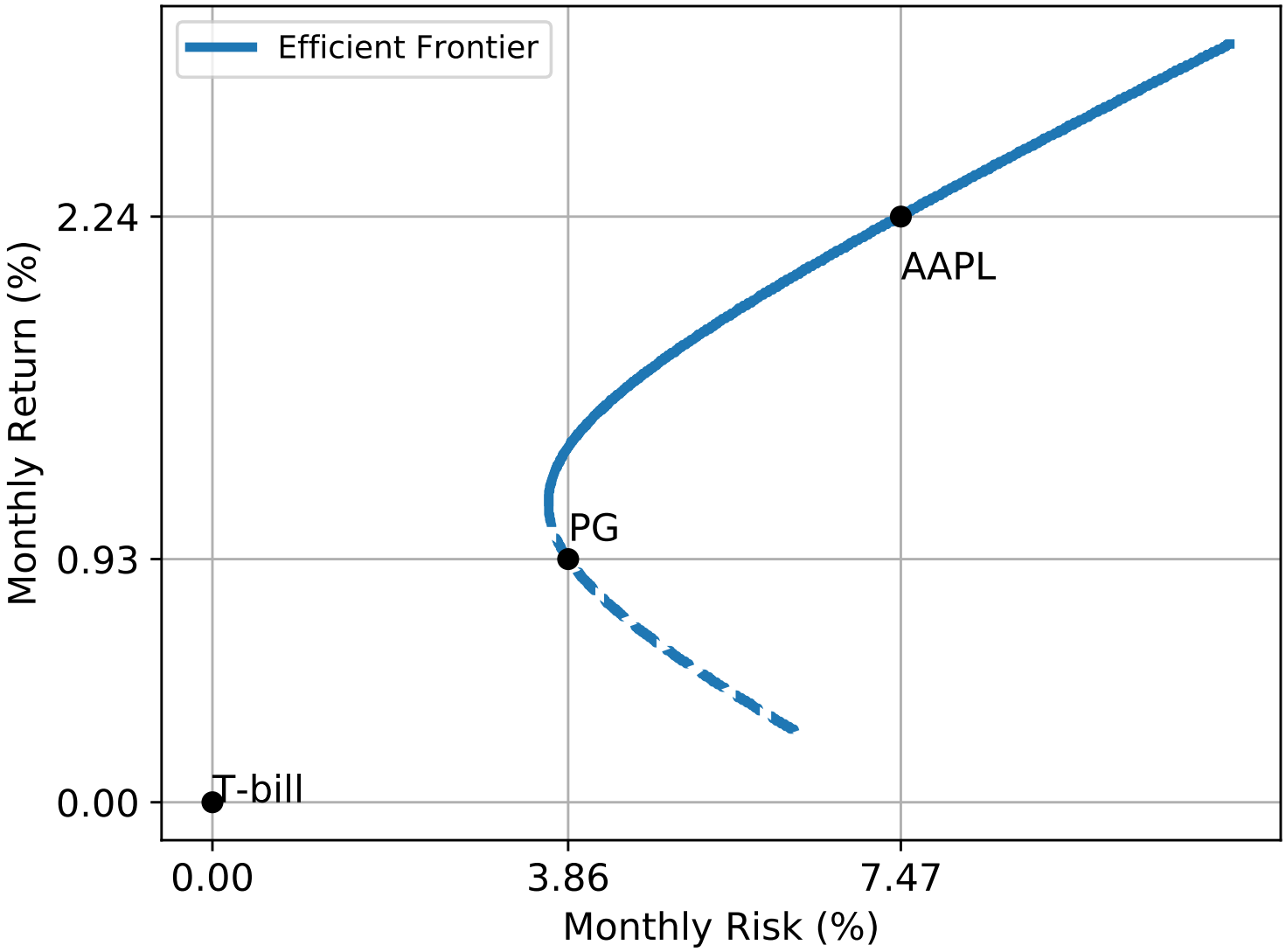
Stock	$E[R]$	$SD[R]$	$SR[R]$
AAPL	2.24%	7.47%	0.3133
PG	0.93%	3.86%	0.2409

- Let's plot this information in risk-return space

Risk-Return Space



Risk-Return Space: Efficient Frontier



Risk-Return Space: Efficient Frontier

- For two stocks, the efficient frontier is the graph of the risks and returns of each portfolio containing the two stocks
- For example, a portfolio that is 100% in Apple and 0% in P&G corresponds to the point labelled AAPL; a portfolio that is 0% in Apple and 110% in P&G corresponds to the point labelled PG; a portfolio that is 50% in Apple and 50% in P&G corresponds to a point along the blue line between AAPL and PG; a portfolio that is 150% in Apple and -50% in P&G corresponds to a point along the blue line north-east of AAPL
- Portfolios associated with points along the dashed line are inefficient, in the sense that for each such portfolio (associated with a point on the bottom of the graph), one can find a portfolio with the same risk but a higher return (associated with a point on the top of the graph)

Moral #4

Suppose you construct a portfolio that is $+150\%$ in AAPL and -50% in PG. Discuss the sequence of cash flows when you initiate the position.

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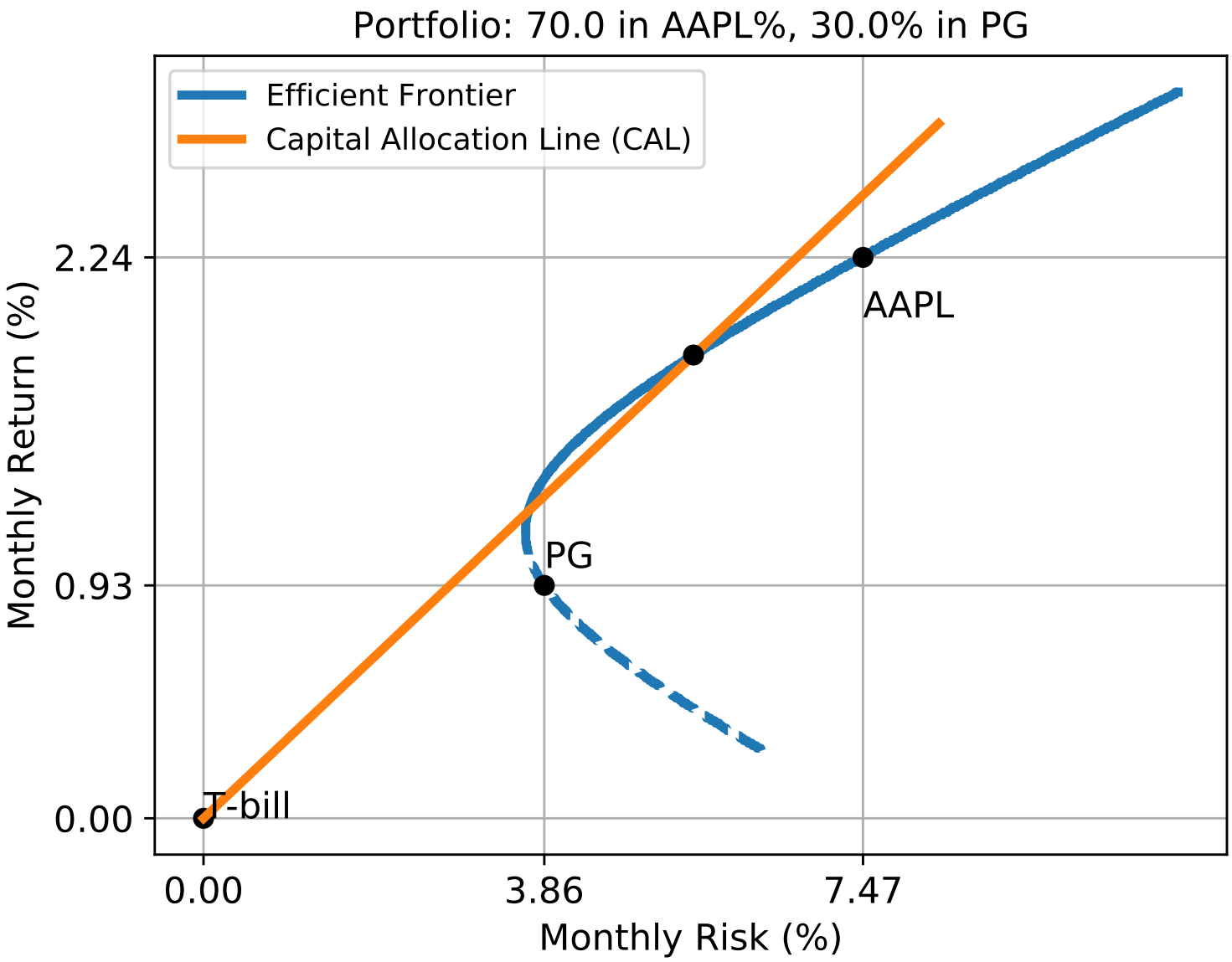
Suppose you construct a portfolio that is $+150\%$ in AAPL and -50% in PG. Discuss the sequence of cash flows when you initiate the position.

Suppose you have 10k USD of capital. Short PG. You collect 5k USD. Together with your 10k USD, buy 15k of AAPL.

Risk-Return Space: Capital Allocation Line (CAL)



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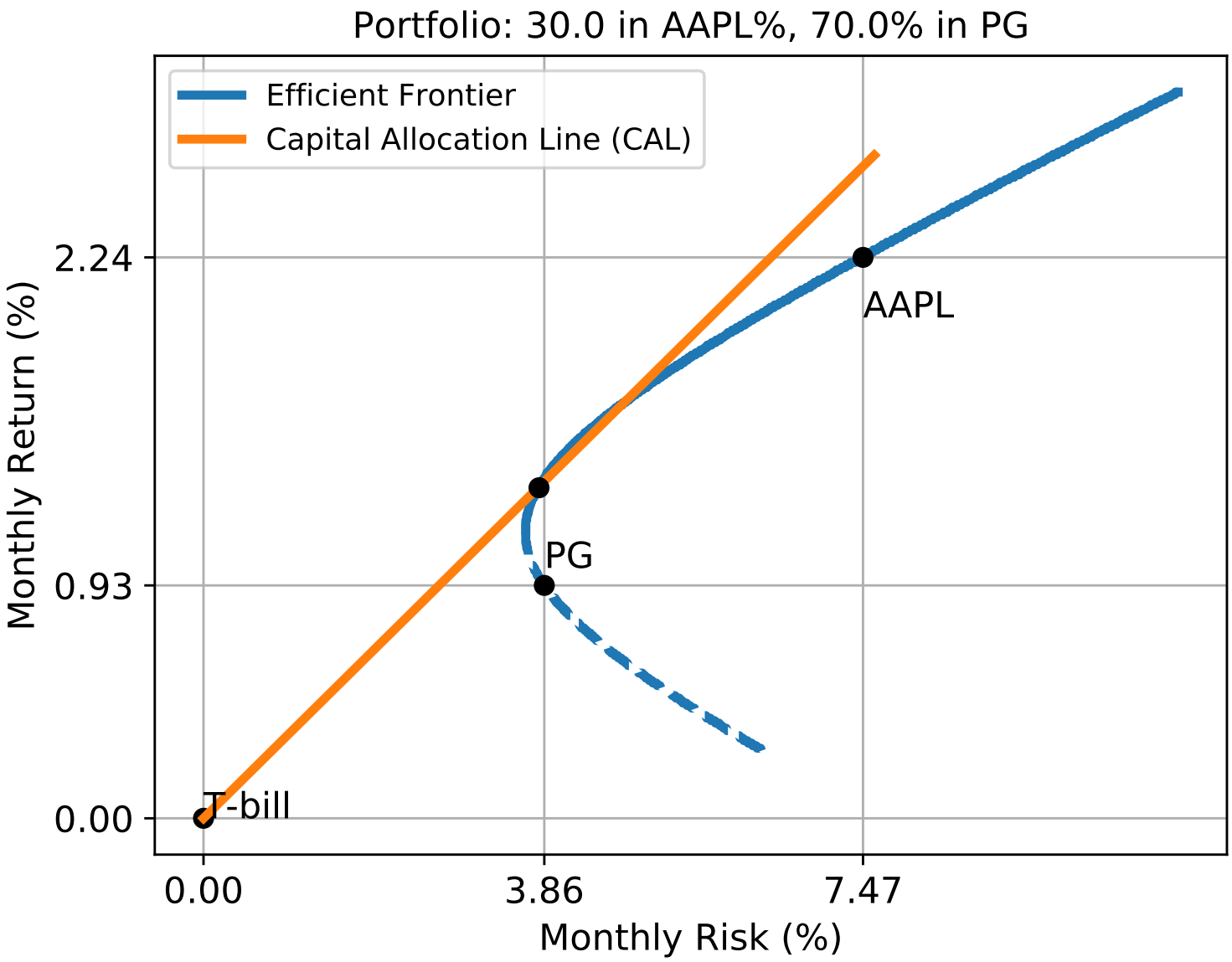


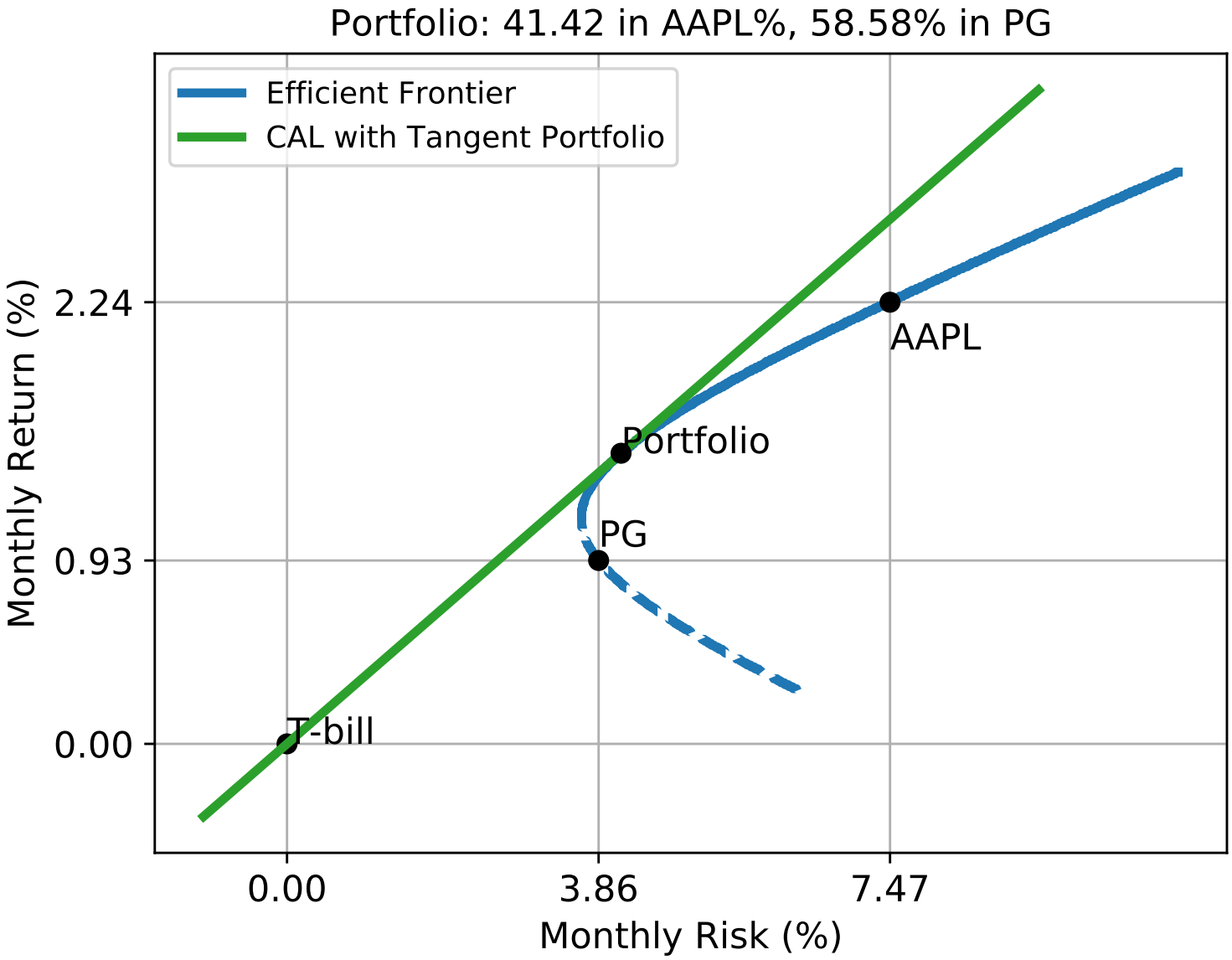
Risk-Return Space: Capital Allocation Line (CAL)

You can use the formula for the volatility of a two stock portfolio with one "stock" being T and the other stock being the risk free. You'll find that

$$\mathrm{E}[R_P] = x\mathrm{E}[rR_T] + (1-x)R_F \tag{4}$$

$$\mathrm{SD}[R_P] = x\mathrm{SD}[R_P] \tag{5}$$

Risk-Return Space: Tangency Portfolio



Risk-Return Space: Capital Allocation Line (CAL)

- The Capital Allocation Line (CAL) consists of the risks and returns of each portfolio containing the risk-free asset and (a portfolio of) the two stocks
- We can construct CALs for any portfolio: we have constructed CALs (the orange lines) for portfolios that are 70% in Apple, 30% in P&G; 70% in Apple, 30% in P&G)
- By pushing the CAL as far north-west as we can while still intersecting the risk-free asset and the efficient frontier, we obtain the tangency portfolio and its CAL (the green line)
- It so happens that this portfolio is 41.42% in Apple and 58.58% in P&G

Tangency Portfolio

- Among all CALs, the one that includes the tangency portfolio has the highest Sharpe Ratio
- To compute the weights of the tangency portfolio, define

$$\widetilde{x}_A = \mathrm{SD}[R_B](\mathrm{SR}[R_A] - \mathrm{SR}[R_B]\mathrm{Corr}[R_A, R_B])$$
$$\widetilde{x}_B = \mathrm{SD}[R_A](\mathrm{SR}[R_B] - \mathrm{SR}[R_A]\mathrm{Corr}[R_A, R_B])$$

The weights of the tangency portfolio are

$$x_A = \frac{\widetilde{x}_A}{\widetilde{x}_A + \widetilde{x}_B}$$
$$x_B = \frac{\widetilde{x}_B}{\widetilde{x}_A + \widetilde{x}_B}$$

Many Stocks

- Thus far, we have studied portfolios consisting of two stocks
- What about portfolios consisting of more than two stocks?
- Consider $N \geq 2$ stocks, $1, 2, \ldots, N$
- The general problem is to choose weights $x_1, x_2, \ldots, x_N$ to minimize the portfolio variance

$$\mathrm{Var}[R_P] = \mathrm{Var}[x_1 R_1 + x_2 R_2 + \cdots + x_N R_N]$$

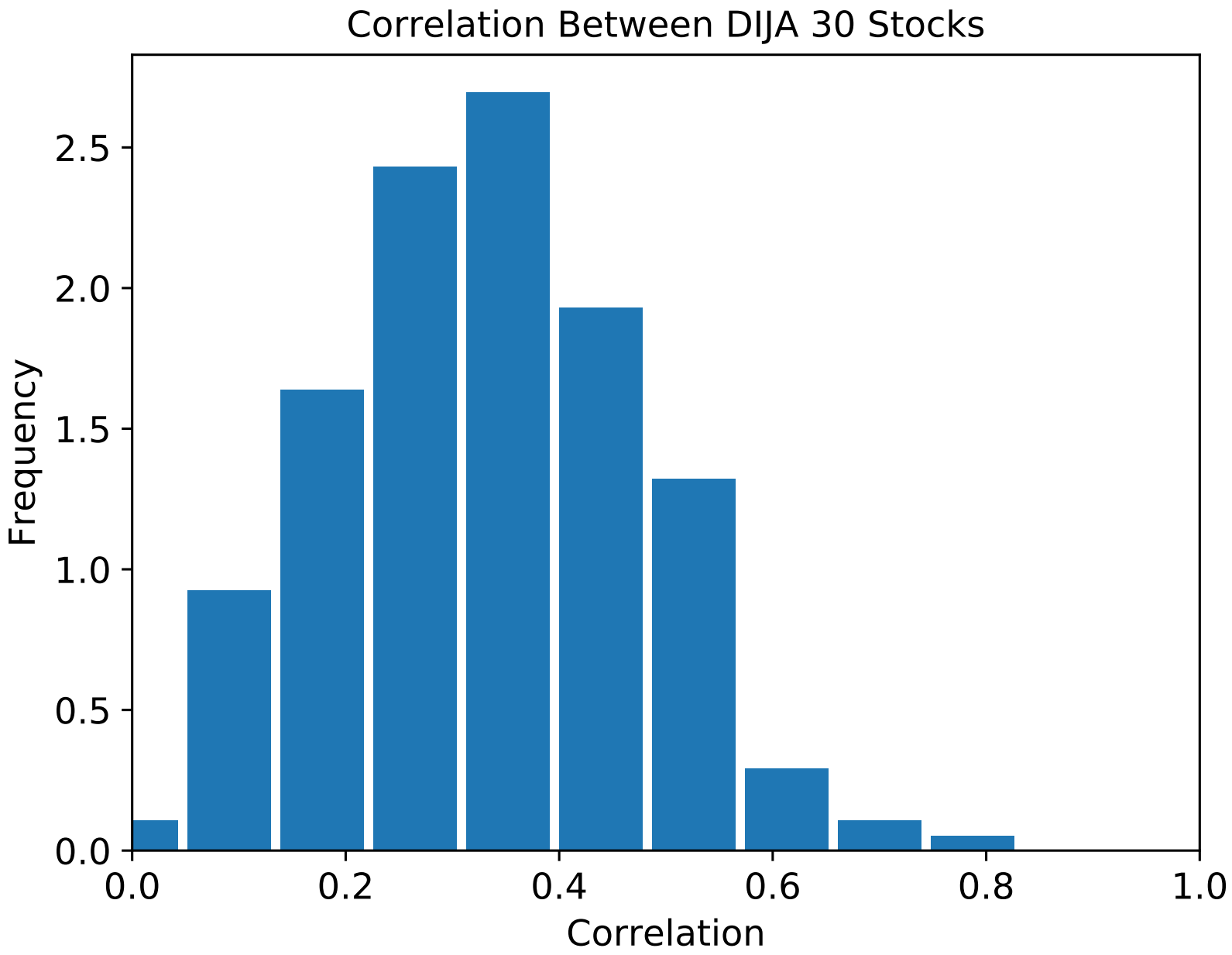
given an expected return μ

$$\mu = \mathrm{E}[R_P] = \mathrm{E}[x_1 R_1 + x_2 R_2 + \cdots + x_N R_N]$$

and $x_1 + x_2 + \cdots + x_N = 1$

- One can solve this problem using *quadratic programming*

Correlations



Special Cases

1. If the stocks are uncorrelated, the portfolio has mean and variance

$$\mathrm{E}[R_P] = x_1\mathrm{E}[R_1] + x_2\mathrm{E}[R_2] + \cdots + x_N\mathrm{E}[R_N] \tag{6}$$

$$\mathrm{Var}[R_P] = x_1^2\mathrm{Var}[R_1] + x_2^2\mathrm{Var}[R_2] + \cdots + x_N^2\mathrm{Var}[R_N] \tag{7}$$

2. An equally weighted portfolio of N independent, identically distributed stocks with mean μ, variance σ^2, and correlation ρ has mean and variance

$$\mathrm{E}[R_P] = \mu \tag{8}$$

$$\mathrm{Var}[R_P] = \sigma^2\left(\frac{1}{N} + \left(1 - \frac{1}{N}\right)\rho\right) \tag{9}$$

Note that $\mathrm{Var}[R_P] \to \rho\sigma^2$ as $N \to \infty$

Example 5

The average standard deviation and correlation of stocks in the Dow Jones Industrial Average are 5.72% and 0.33 respectively. What is the standard deviation of an equally-weighted portfolio of 2 stocks? 4 stocks? 8 stocks?

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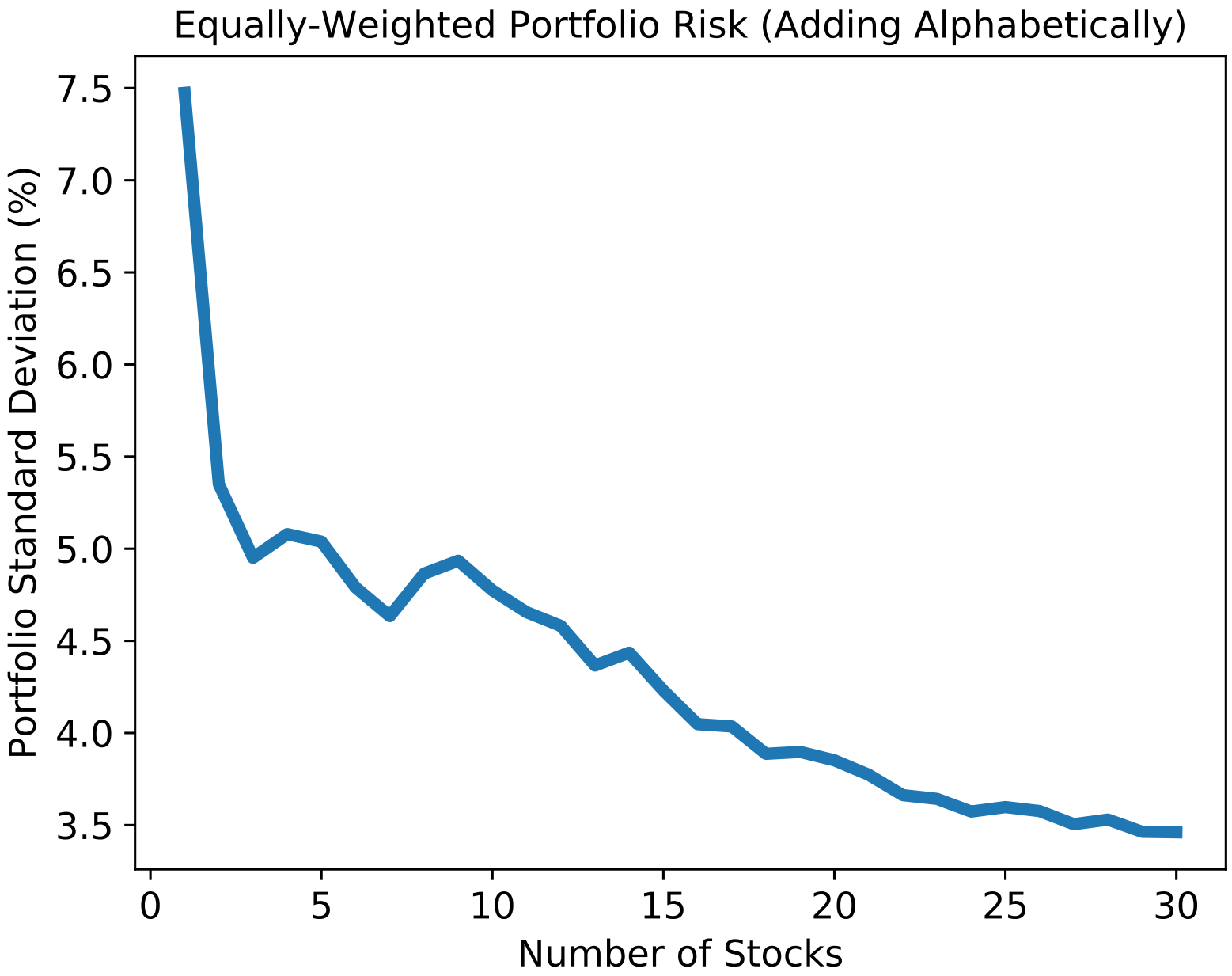
$$\mathrm{SD}[R_{2\ \mathrm{Stocks}}] = \sqrt{(.0572)^2\left(\frac{1}{2} + \left(1 - \frac{1}{2}\right)(0.33)\right)} = 4.6645\%$$

$$\mathrm{SD}[R_{4\ \mathrm{Stocks}}] = \sqrt{(.0572)^2\left(\frac{1}{4} + \left(1 - \frac{1}{4}\right)(0.33)\right)} = 4.0345\%$$

$$\mathrm{SD}[R_{8\ \mathrm{Stocks}}] = \sqrt{(.0572)^2\left(\frac{1}{8} + \left(1 - \frac{1}{8}\right)(0.33)\right)} = 3.6793\%$$

The Value of Diversification